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# PHY1001 Mechanics

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# Teaching and Learning Feedback Questionnaire



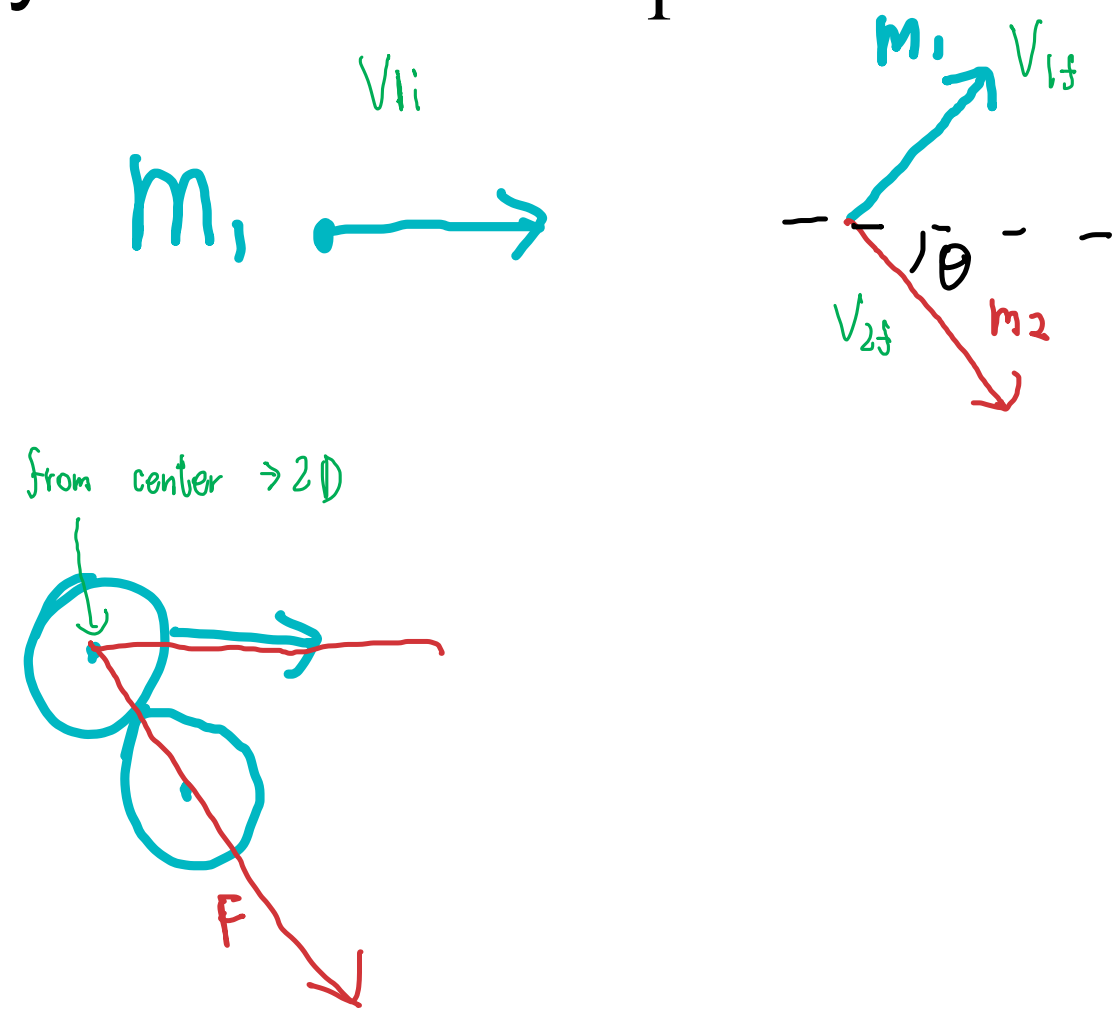
- The purpose of this questionnaire is to evaluate and enhance the teaching and learning process.
- Your personal information will remain confidential.

# Collision in 2D

- Conservation of momentum:  $\vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_{1f} + \vec{P}_{2f}$ 
  - X direction:  $P_{1i,x} + P_{2i,x} = P_{1f,x} + P_{2f,x}$
  - Y direction:  $P_{1i,y} + P_{2i,y} = P_{1f,y} + P_{2f,y}$
- Conservation of kinetic energy (if elastic):  $K_{1i} + K_{2i} = K_{1f} + K_{2f}$

# Exercise

A particle of mass  $m_1$  scatters elastically with a particle of mass  $m_2$ , and  $m_2$  recoils with an angle  $\theta$  w.r.t to the original velocity direction of  $m_1$ . The velocity of  $m_1$  before scattering is  $v_{1i}$ . Find the recoil velocity  $v_{2f}$  of  $m_2$ .



$$P_x: \begin{cases} m_1 v_{1i} = m_2 v_{2f} \cos \theta + m_1 v_{1f} \cos \varphi \\ m_1 v_{1i} \sin \varphi = m_2 v_{2f} \sin \theta \end{cases}$$

$$K: \frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

$$\begin{aligned} \frac{1}{2} m_1 v_{1i}^2 &= \frac{1}{2} m_1 (v_{1f} \sin \varphi)^2 + \frac{1}{2} m_1 (v_{1f} \cos \varphi)^2 + \frac{1}{2} m_2 v_{2f}^2 \\ &= \frac{1}{2} \frac{(m_2 v_{2f} \sin \theta)^2}{m_1} + \frac{1}{2} \frac{(m_1 v_{1i} - m_2 v_{2f} \cos \theta)^2}{m_1} + \frac{1}{2} m_2 v_{2f}^2 \end{aligned}$$

$$m_1^2 v_{1i}^2 = m_2^2 v_{2f}^2 \sin^2 \theta + m_1^2 v_{1i}^2 + m_2^2 v_{2f}^2 \cos^2 \theta - 2m_1 m_2 v_{1i} v_{2f} \cos \theta + m_1 m_2 v_{2f}^2$$

$$v_{2f} = \frac{2m_1 v_{1i} \cos \theta}{m_1 + m_2}$$

In 1D  $\rightarrow \theta = 0 \rightarrow \cos \theta = 1$

$$v_{2f} = \frac{2m_1 v_{1i}}{m_1 + m_2}$$

# Exercise

Dark matter scatters with xenon atoms:

- $m_1 = 50$  GeV (Dark matter candidate, Weakly Interacting Massive Particle (WIMP))
- $m_2 = 131$  GeV (xenon atom)  $\rightarrow$  use for measure wind
- $v_{1i} = 300$  km/s  $= 0.001$  c

$$V_{2f} = \frac{2m_1 V_{1i} \cos \theta}{m_1 + m_2}$$

$$= \frac{2 \cdot 50 \cdot 10^{-3} \text{ c} \cdot \cos \theta}{131 + 50}$$

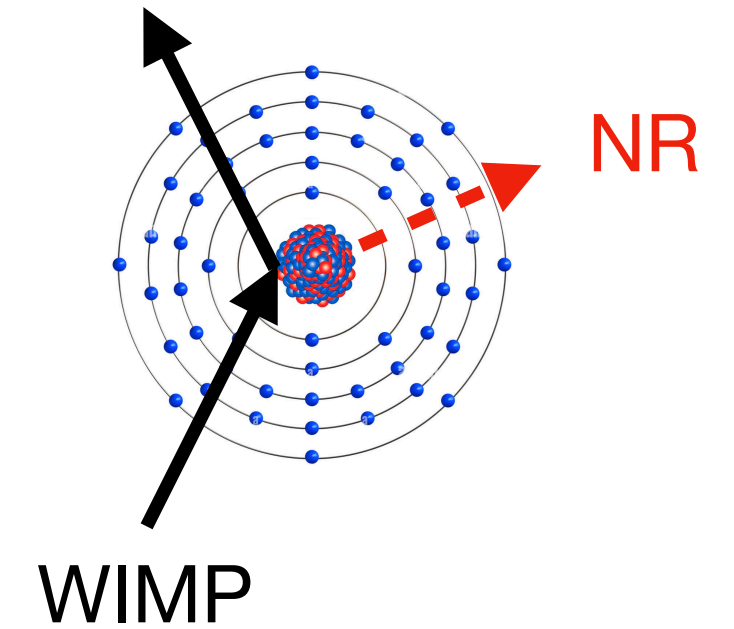
$$= \frac{100}{181} \times 10^{-3} \text{ c} \cdot \cos \theta$$

$$K = \frac{1}{2} m_2 V_{2f}^2$$

$$= \frac{1}{2} \cdot 131 \text{ GeV}/c^2 \left( \frac{100}{181} \cdot 10^{-3} \text{ c} \cdot \cos \theta \right)^2$$

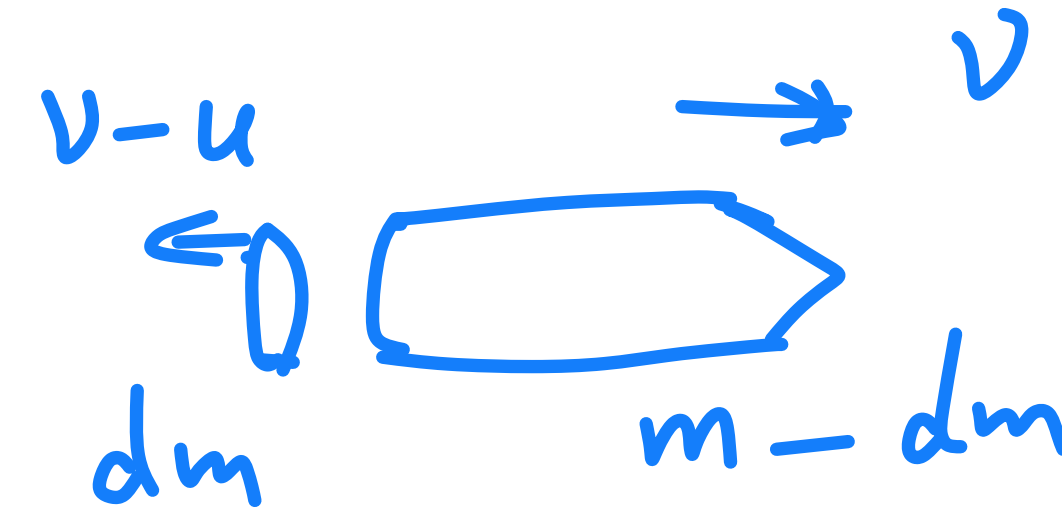
$$= \frac{1}{2} \cdot 131 \left( \frac{100}{181} \right)^2 \cdot 10^{-6} \text{ c}^2 \cdot \frac{\text{GeV}}{\text{c}^2} \cdot \cos^2 \theta$$

$$= 20 \cos^2 \theta \text{ keV}$$



# Rocket Propulsion: Varying Mass

- Fuel is rejected from the rocket with a relative velocity  $u$  ( $v_{rel}$  in some textbook) to the rocket.
- The momentum of the system of the rejected fuel and the rocket is conserved
- First rocket equation:  $Ru = ma$ , where
  - $R$  is the fuel consumption rate in kg/s
  - $m$  is the instantaneous mass of the rocket
  - $a$  is the instantaneous acceleration



$$mv = (m-dm)(v+dv) - dm(v-u)$$

$$= mv - vdm + m dv - \boxed{dm dv} + vdm - udm$$

$$m dv = u dm$$

$$m dv = U R dt$$

$$m \frac{dv}{dt} = RU$$

$$\boxed{F = ma = RU} \quad \text{first rocket equation}$$

(Thrust)

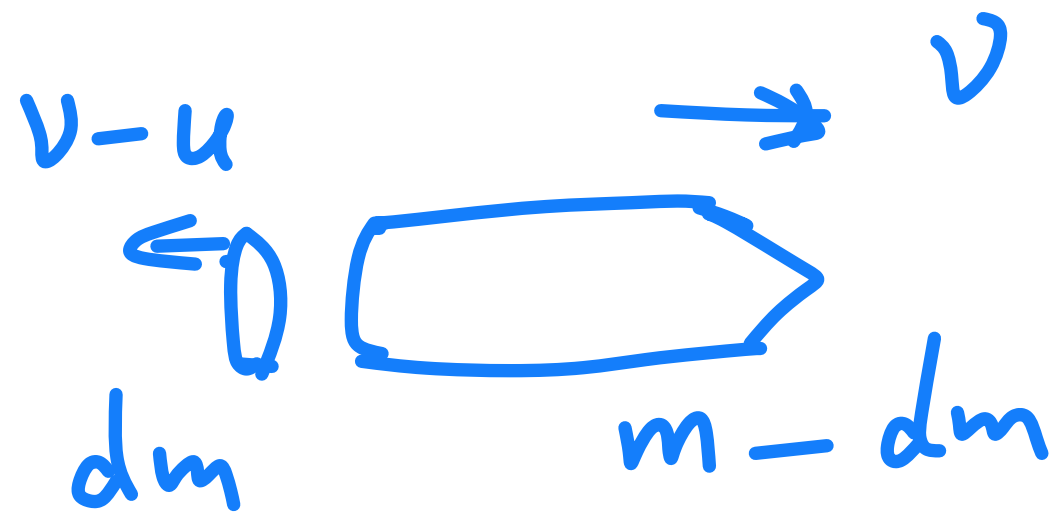
$$R = \frac{dm}{dt}$$

$$\begin{aligned} [RU] &= \text{kg/s} \cdot \text{m/s} \\ &= \text{kg} \cdot \text{m/s}^2 \end{aligned}$$

# Rocket Propulsion: Varying Mass

- Second rocket equation:  $v_f - v_i = u \ln \frac{M_i}{M_f}$ , where

- $v_i$ : initial velocity of the rocket
- $v_f$ : final velocity
- $M_i$ : initial mass of the rocket and the fuel
- $M_f$ : final mass of the rocket and the fuel



$$m dv = u dm$$
$$\int_{v_i}^{v_f} dv = - \int_{m_i}^{m_f} u \frac{dm}{m}$$

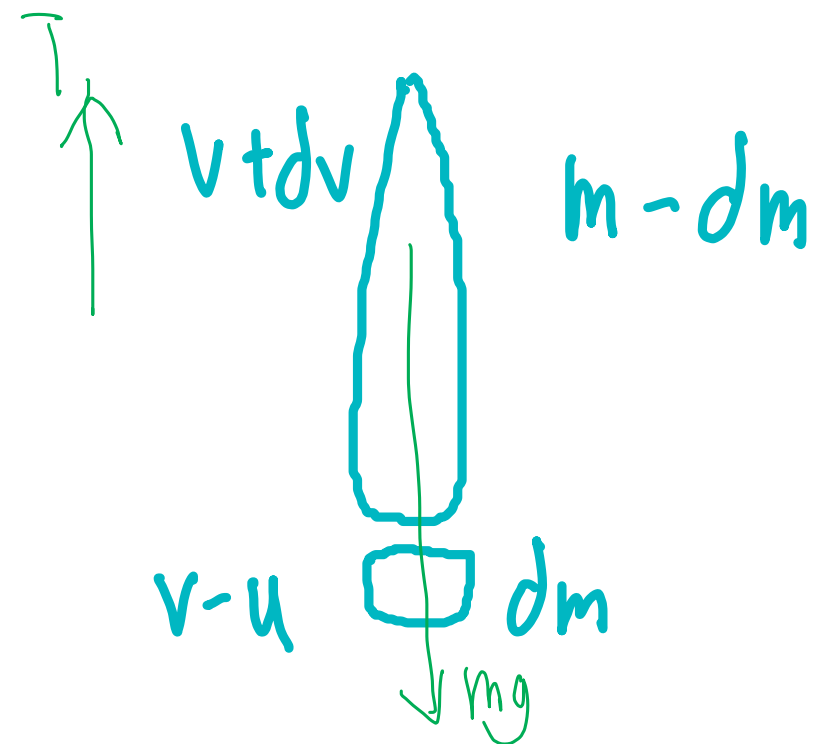
$$\Delta v = v_f - v_i = u \ln \frac{m_i}{m_f}$$

# Exercise

A rocket is set for a vertical launch. The mass of the rocket is  $m_0$  kg and the mass of the fuel is  $m_f$ . The exhaust velocity of the fuel relative to the rocket is  $v_{\text{rel}}$ .

a) Find the minimum consumption rate  $R_{\text{min}}$  to lift off the rocket.

momentum - impulse Th



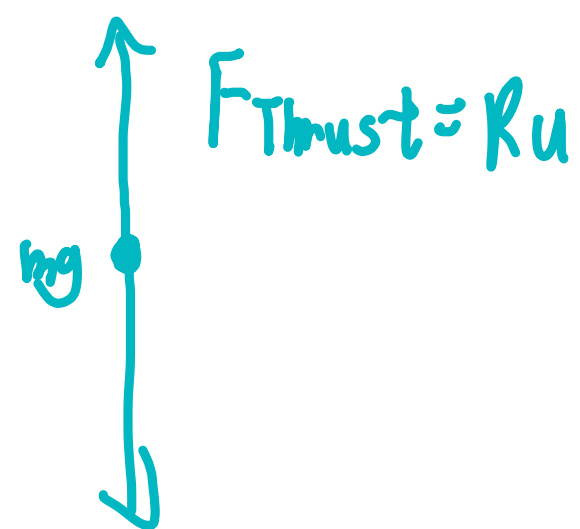
$$J = \Delta P$$

$$\begin{aligned} -mg \cdot dt &= (m-dm)(v+dv) + (v-u)dm - mv \\ &= mv - vdm + m dv + vdm - udm - mv \\ &= m dv - u dm \end{aligned}$$

$$-mg = ma - uR$$

$$Ru - mg = ma \quad \text{must } > 0 \text{, } \times \text{gravity}$$

$$R > \frac{mg}{u}$$



# Exercise

A rocket is set for a vertical launch. The mass of the rocket is  $m_0$  kg and the mass of the fuel is  $m_f$ . The exhaust velocity of the fuel relative to the rocket is  $v_{\text{rel}}$ .

b) Assume the consumption rate is fixed at  $R_0$ . What is the final velocity  $v$  when the fuel is completely consumed?



$$J = \Delta p$$

$$-mg dt = m dv - u dm$$

$$dm = R dt \Rightarrow dt = \frac{dm}{R}$$

$$-mg \frac{dm}{R} = m dv - u dm$$

$$\int_{m_f}^{m_i} \frac{1}{m} \left( u - \frac{mg}{R} \right) dm = \int_{v_i}^{v_f} dv = v_f - v_i$$

$$u \ln \frac{m_i}{m_f} - \frac{g}{R} (m_f - m_i)$$

$$u \ln \frac{m_0 + m_f}{m_0} - \frac{g}{R} \Delta t = v_f$$

↳ conserved!

# Center of Mass

- The center of mass is the mass-weighted center of a system

- mathematically:  $x_{\text{com}} = \frac{\sum_i m_i x_i}{\sum_i m_i}$

- For higher dimensions:  $\vec{r}_{\text{com}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$

- The total momentum of a system  $\vec{P} = M\vec{v}_{\text{com}}$ , where

- $M = \sum_i m_i$

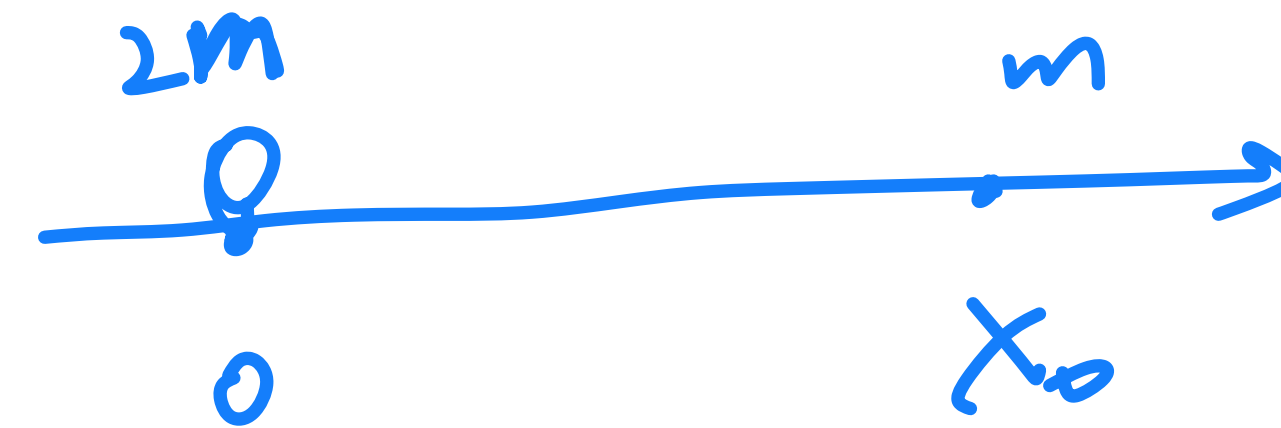
- $\vec{v}_{\text{com}}$  is the velocity of center of mass

- Useful for momentum conservation calculation

# Exercise

A ball of mass  $2m$  and a ball  $m$  are on a horizontal smooth surface. The  $2m$  ball is at the origin, while the  $m$  ball is at  $x = x_0$ , as shown in the plot.

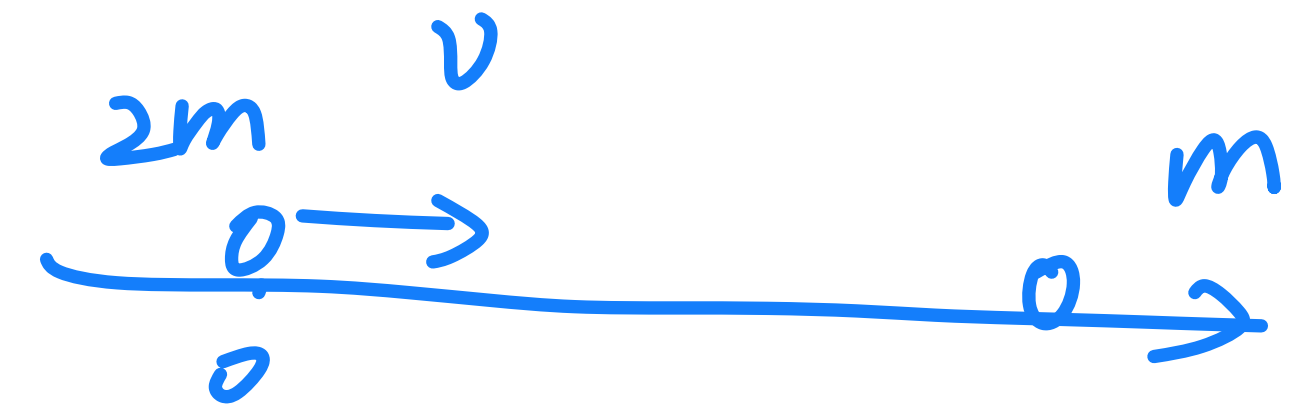
a) Find the center of mass of this two-ball system.



# Exercise

A ball of mass  $2m$  collides elastically with a ball  $m$  on a horizontal smooth surface. The  $2m$  ball has a velocity of  $v$ , while the  $m$  ball is at rest before the collision.

b) Find the velocity of the two balls after the collision using the lab frame.



# Exercise

A ball of mass  $2m$  collides elastically with a ball  $m$  on a horizontal smooth surface. The  $2m$  ball has a velocity of  $v$ , while the  $m$  ball is at rest before the collision.

c) Find the velocity of the two balls after the collision using the frame of center of mass.

